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Java SE 27 & JDK 27

Interface WritableByteChannel

All Superinterfaces:

AutoCloseable, Channel, Closeable

All Known Subinterfaces:

ByteChannel, GatheringByteChannel, SeekableByteChannel

All Known Implementing Classes:

DatagramChannel, FileChannel, Pipe.SinkChannel, SocketChannel

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```
public interface WritableByteChannel
extends Channel
```

A channel that can write bytes.

Only one write operation upon a writable channel may be in progress at any given time. If one thread initiates a write operation upon a channel then any other thread that attempts to initiate another write operation will block until the first operation is complete. Whether or not other kinds of I/O operations may proceed concurrently with a write operation depends upon the type of the channel.

Since:

1.4

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
int	write(ByteBuffer src)	Writes a sequence of bytes to this channel from the given buffer.

Methods declared in interface Channel

close, isOpen

Method Details

write

```
int write(ByteBuffer src)
   throws IOException
```

Writes a sequence of bytes to this channel from the given buffer.

An attempt is made to write up to *r* bytes to the channel, where *r* is the number of bytes remaining in the buffer, that is, `src.remaining()`, at the moment this method is invoked.

Suppose that a byte sequence of length *n* is written, where `0` <= *n* <= *r*. This byte sequence will be transferred from the buffer starting at index *p*, where *p* is the buffer's position at the moment this method is invoked; the index of the last byte written will be *p* + *n* - 1. Upon return the buffer's position will be equal to *p* + *n*; its limit will not have changed.

For many types of channels, a write operation will return only after writing all of the *r* requested bytes. Some types of channels, depending upon their state, may write only some of the bytes or possibly none at all. A socket channel in non-blocking mode, for example, cannot write any more bytes than are free in the socket's output buffer. The write method may need to be invoked more than once to ensure that all remaining bytes are written.

This method may be invoked at any time. If another thread has already initiated a write operation upon this channel, however, then an invocation of this method will block until the first operation is complete.

Parameters:

`src` - The buffer from which bytes are to be retrieved

Returns:

The number of bytes written, possibly zero

Throws:

`NonWritableChannelException` - If this channel was not opened for writing

`ClosedChannelException` - If this channel is closed

`AsynchronousCloseException` - If another thread closes this channel while the write operation is in progress

`ClosedByInterruptException` - If another thread interrupts the current thread while the write operation is in progress, thereby closing the channel and setting the current thread's interrupted status

`IOException` - If some other I/O error occurs

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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